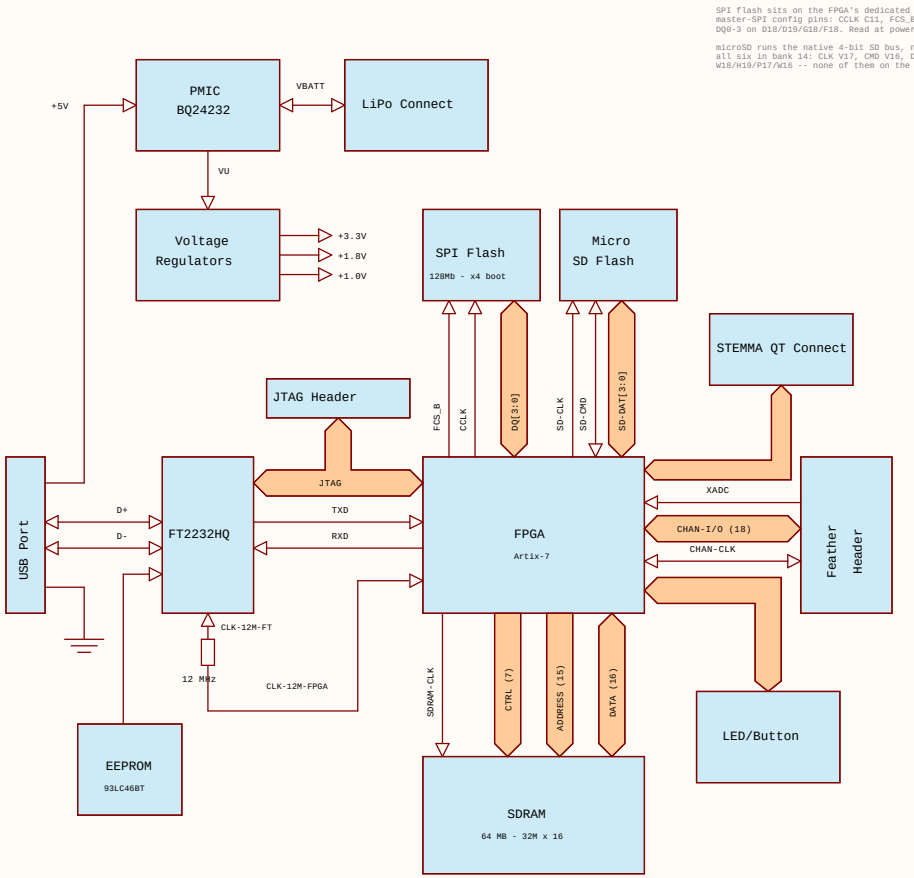
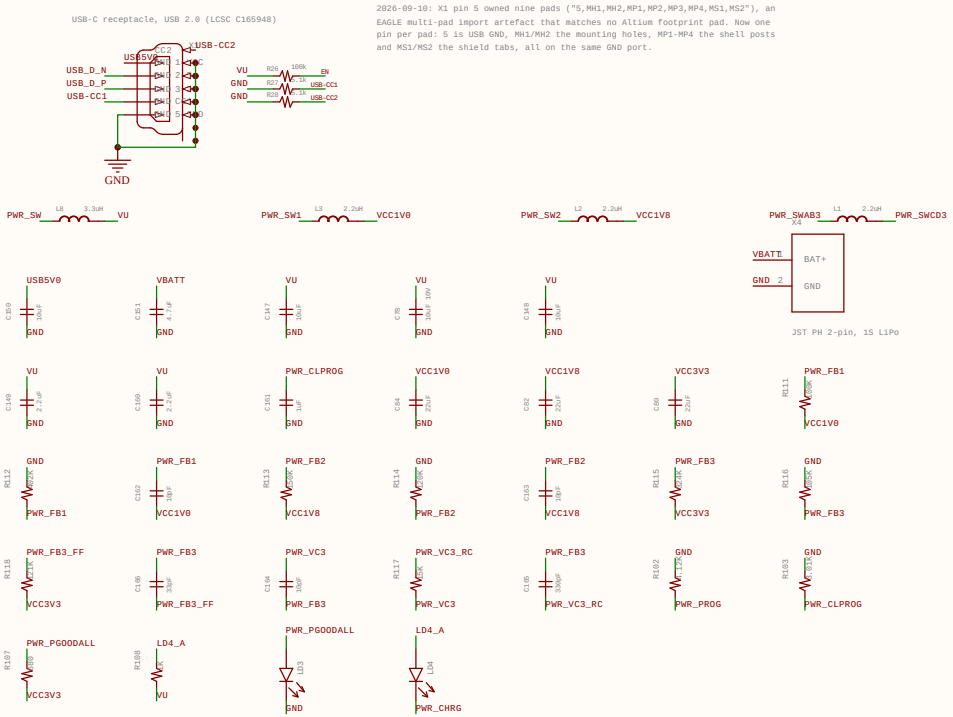


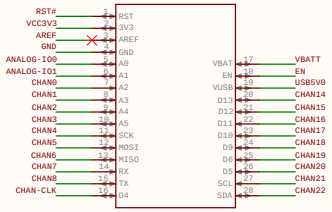
Block Diagram			
PROJECT:	asar_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR:	Christian Tambe		Sheet: 1



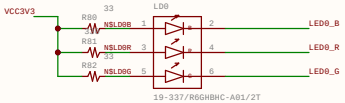
Power Supplies			
PROJECT:	asar_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR:	Christian Tambe		Sheet: 2

X2: Feather headers, a 1x16 and a 1x12 strip, 0.1 in pitch, rows 0.8 in (20.32 mm) apart.
Pads 1-16 form the 16-pin strip, 17-28 the 12-pin strip; nets keep the Zulu A7 CHANn names.
16-PIN: 1 RST (RST=: PROGRAM_B V10 and the reset button), 2 3V3 (VCC3V3), 3 AREF (no load),
  4 GND, 5 A0 and 6 A1 (ANALOG-IO0/1 -> the XADC dividers on AD4 and AD12),
  7-10 A2-A5 = CHAN0-CHAN3, 11 SCK = CHAN4, 12 MOSI = CHAN5, 13 MISO = CHAN6, 14 RX = CHAN7,
  15 TX = CHAN8, 16 D4 = CHAN-CLK (ball P18, clock-capable).
12-PIN: 17 VBAT (VBATT, the LiPo on X4), 18 EN (EN pins of U5-U7), 19 VUSB (USB5V0 from X1),
  20-24 D13, D12, D11, D10, D9 = CHAN14-CHAN18, 25 D6 = CHAN19, 26 D5 = CHAN20,
  27 SCL = CHAN21, 28 SDA = CHAN22.
FPGA balls: vivado/asar_feather_pins.xdc and the pin table in README.md.

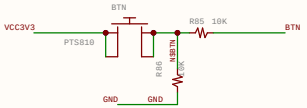
Feather headers: 1x16 + 1x12, 0.1 in



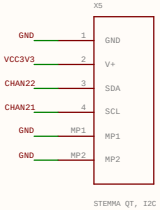
2026-09-10: BTN pads 1-2 are one switch terminal and 3-4 the other. They were one pin each ("1,2" and "3,4"), an EAGLE multi-pad import artefact that matches no Altium footprint pad; now four pins, each pad on its partner's net.



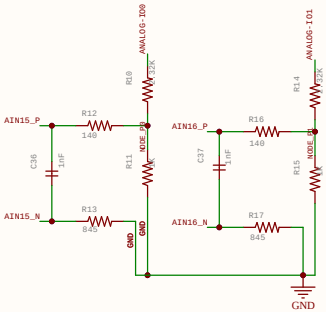








STEMMA QT, I2C

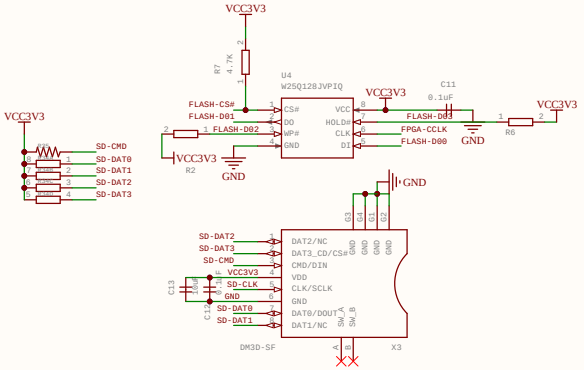


2026, Christian Tambe. Some rights reserved.

Licensed under a Creative Commons
Attribution-Share-Alike Unported License.
See creativecommons.org/licenses/by-sa/3.0/

General IO			
PROJECT: asar_a7		REV:	9/2/2026 1:29:21 PM
AUTHOR: Christian Tambe			Sheet: 3

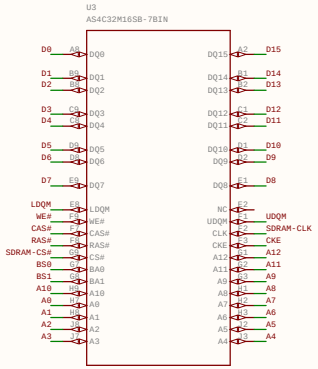
MASTER SPI BOOT (x4). M2:M1:M0 = 0:0:1 -- R20 pulls M0 to VCC3V3, R21/R22 pull M1/M2 to GND -- so at power-up the FPGA drives CCLK itself and clocks its bitstream out of U4 with no USB and no host present. That mode can only use the dedicated configuration pins: CCLK C11, FCS_B K19, D00_MOSI D18, D01_DIN D19, D02 G18, D03 F18. No other ball will do, which is why SDRAM A4 had to move off K19 (it is now on R3, bank 34). After DONE the six pins revert to ordinary user I/O. R2/R6 keep WP#/HOLD# high through the x1 command phase and through the float window before the FPGA drives D02/D03 (PUDC_B is tied high, so the FPGA has no internal pull-ups). R7 keeps FCS_B deselected. Vivado: set_property CONFIG_MODE SPIx4 [current_design] set_property BITSTREAM.CONFIG.SPI_BUSWIDTH 4 [current_design] The microSD is deliberately NOT on this bus any more: an inserted card sits in native SD mode watching bitstream traffic go past on what it believes is CMD, and one 48-bit window with a valid CRC7 is enough to make it answer and drive the line the FPGA is reading. X3 has its own CLK/CMD/DAT0 on V17/V16/W18 (bank 14), clear of the six master-SPI pins.

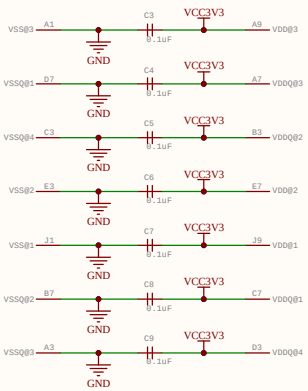


microSD runs the NATIVE 4-BIT SD BUS, not SPI. CLK, CMD and DAT0-DAT3 -- six signals, all in bank 14: CLK V17, CMD V16, DAT0 W18, DAT1 H19, DAT2 P17, DAT3 W16. DAT3 is the old SPI chip-select, which needs no new pin. R34/R35 are the 4.7k pull-ups the SD spec requires on CMD and DAT0-DAT3. R9 (100R on DAT0 only) was removed: one damped line out of four is a skew, not termination.

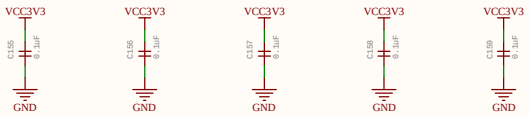
2026-09-06: X3 is now the Hirose DM3D-SF push-pull socket (HRS 609-0025-8), DM3 catalog p9. Same 8-contact pinout; shell pads G1-G4 to GND; card-detect switch A/B (closed with card) left open. Needs its own footprint on the PCB.

U3 = AS4C32M16SB-7BIN: 512Mbit (32M x 16) = 64MB, 54-ball FBGA, 8 x 8 mm, 0.8 mm ball pitch. The column address is A0-A9 (1024 columns): the memory controller must be set to 10 column bits, or it reaches only half the array. Row A0-A12, banks BA0/BA1, refresh 8192 cycles / 64 ms. -7BIN: 143 MHz max at CL3, tRCD/tRP 21, tRC 63 ns, TA -40 to 85 C (industrial), IDD1 max 110 mA.

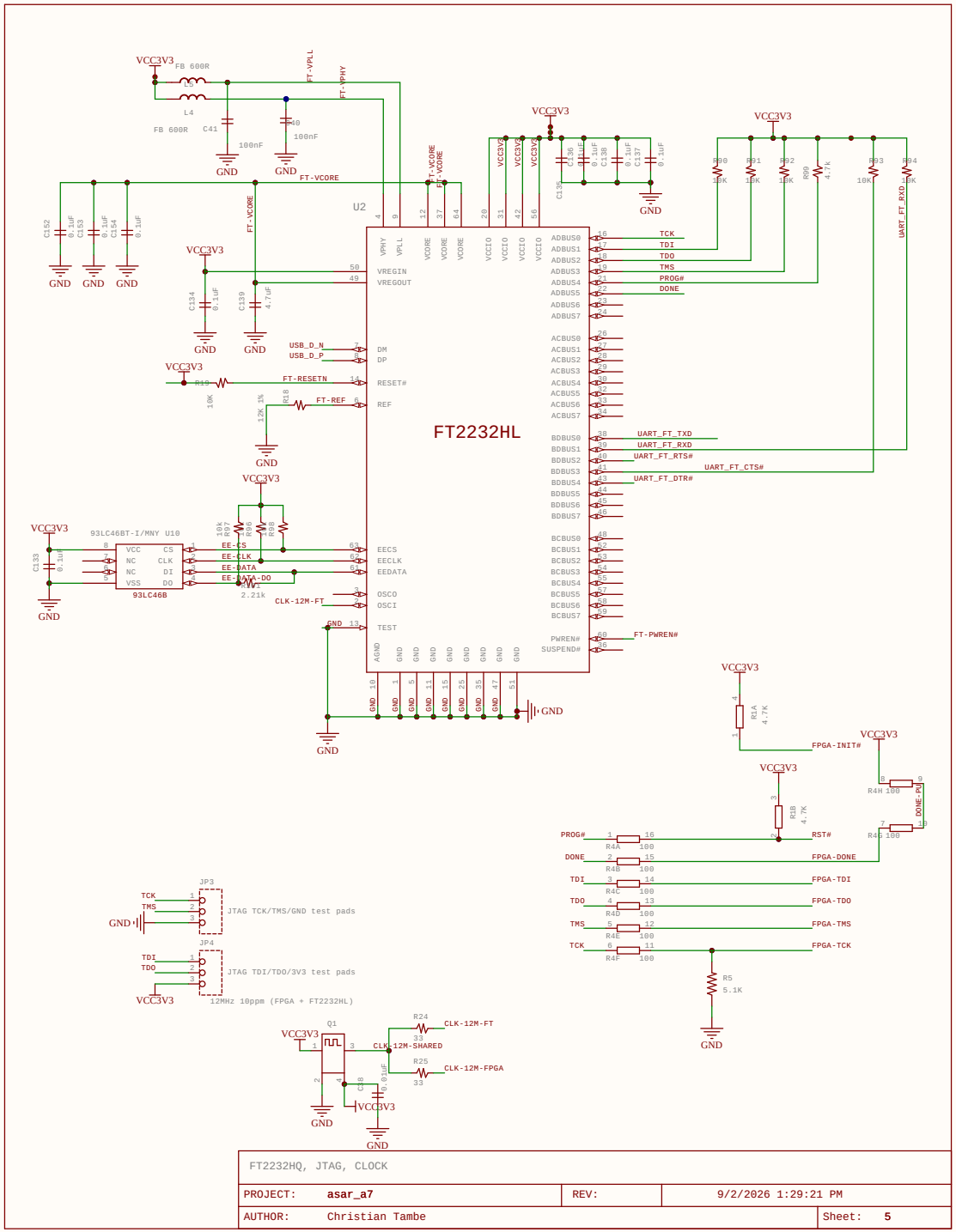


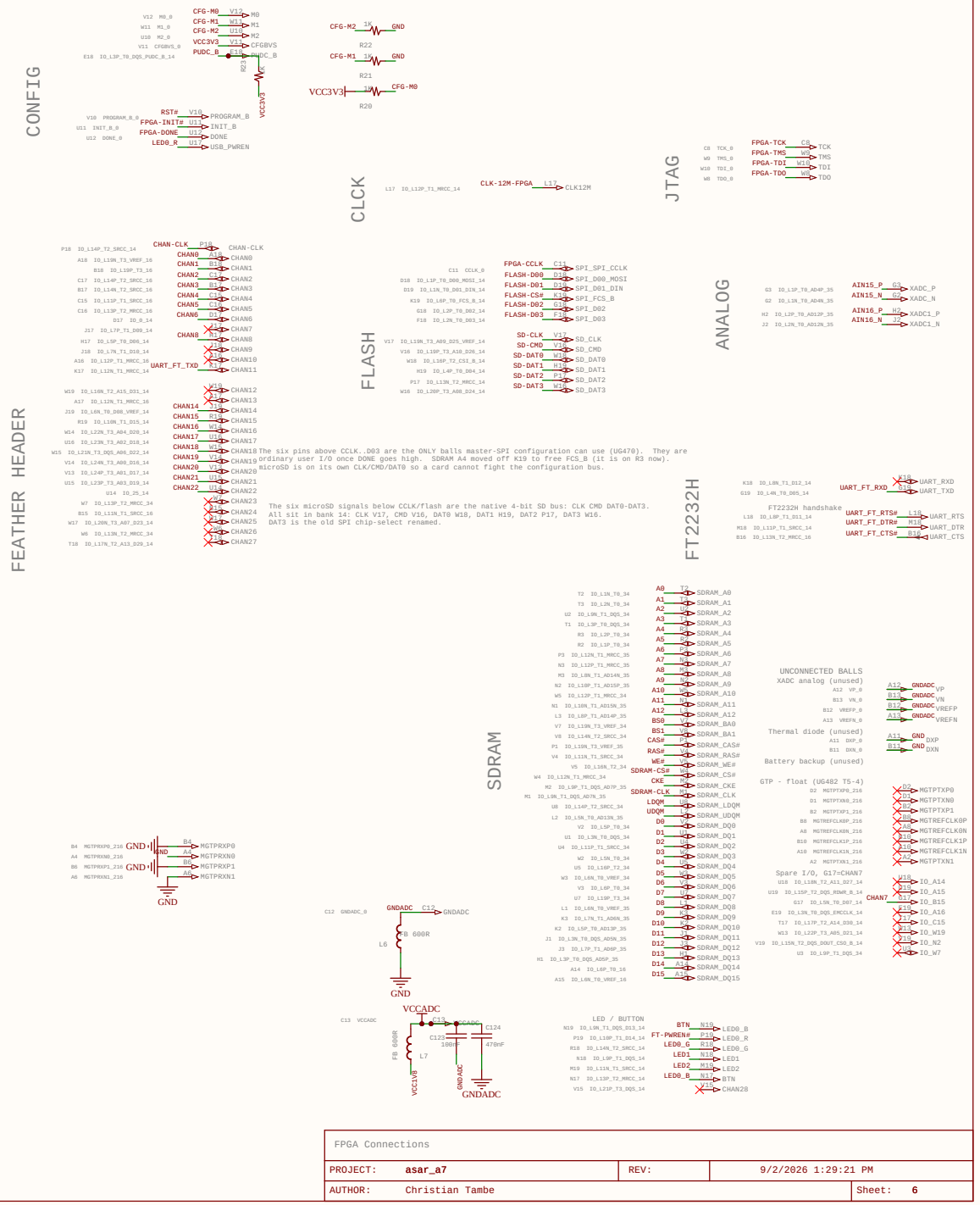


U3 DECOUPLING -- U3 is the 54-ball FBGA. Each supply ball has a ground of its own domain at the mirror position in its row: VDD A9/E7/J9 to VSS A1/E3/J1, VDDQ A7/B3/C7/D3 to VSSQ A3/B7/C3/D7. Alliance isolates VDDQ from VDD for DQ noise immunity; on this board both are VCC3V3 and both grounds are GND. C3-C9, C11 and C12 (0.1 uF) and C13 (10 uF bulk) are on the bottom side under U3. Do NOT split the ground plane.



Memory			
PROJECT: asar_a7		REV:	9/2/2026 1:29:21 PM
AUTHOR: Christian Tambe			Sheet: 4

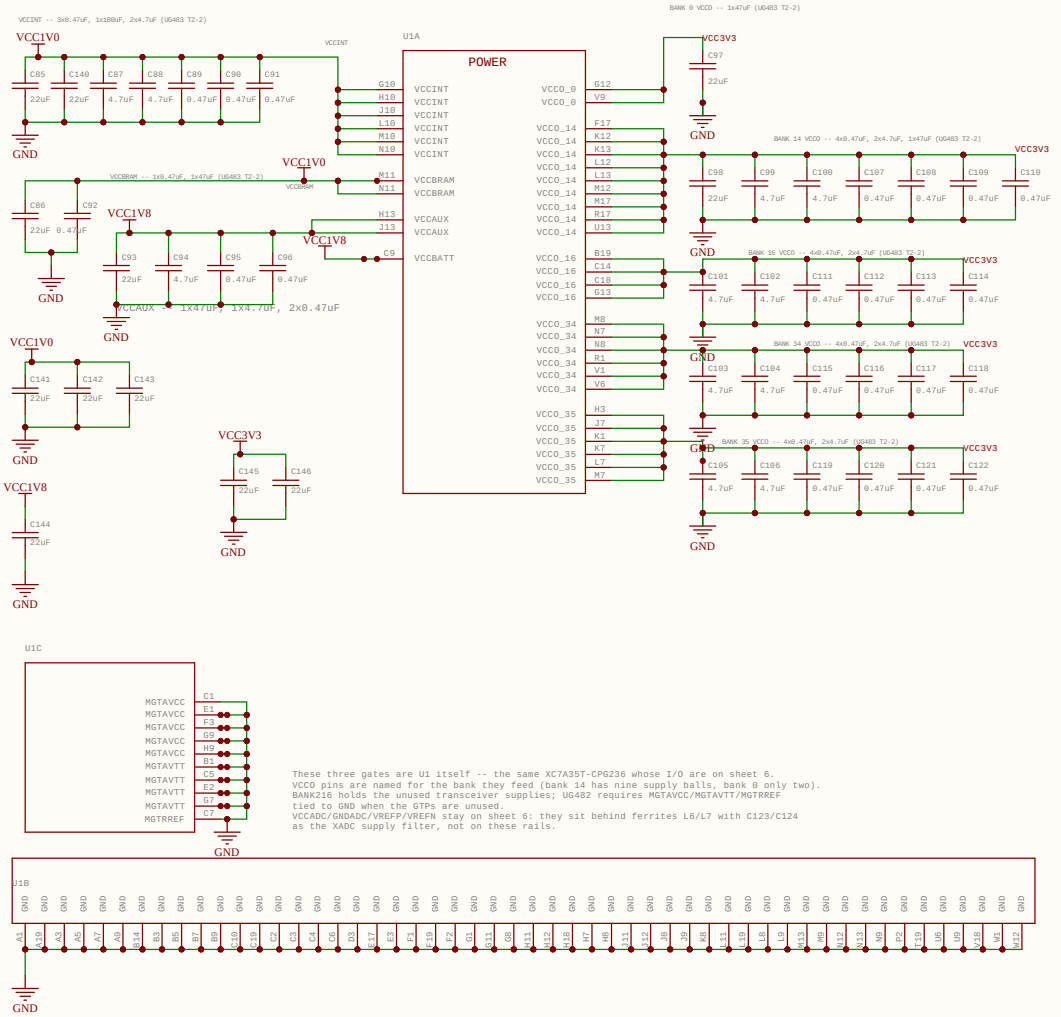

FT2232HL
U2
93LC46B
JTAG TCK/TMS/GND test pads
JTAG TDI/TDO/3V3 test pads
12MHz 10ppm (FPGA + FT2232HL)
FT2232HQ, JTAG, CLOCK
PROJECT: asar_a7
REV:
9/2/2026 1:29:21 PM
AUTHOR: Christian Tambe
Sheet: 5



The six pins above CCLK..D03 are the ONLY balls master-SPI configuration can use (UG470). They are ordinary user I/O once DONE goes high. SDRAM A4 moved off K19 to free FCS_B (it is on R3 now). MicroSD is on its own CLK/CMD/DAT0 so a card cannot fight the configuration bus.

The six microSD signals below CCLK/flash are the native 4-bit SD bus: CLK CMD DAT0-DAT3. All sit in bank 14: CLK V17, CMD V16, DAT0 W18, DAT1 H19, DAT2 P17, DAT3 W16. DAT3 is the old SPI chip-select renamed.

FPGA Connections			
PROJECT: asar_a7		REV:	9/2/2026 1:20:21 PM
AUTHOR: Christian Tambe			Sheet: 6



FPGA POWER			
PROJECT:	asar_a7	REV:	9/2/2026 1:29:21 PM
AUTHOR:	Christian Tambe		Sheet: 7